SIGMIDI Foot Controller v0.1

Bill of materials - one completed controller

Preferred build: **Left**, with electronics on the left and modular extensions to the right. The Right variant uses the same purchased electrical parts; select its matching printed files. Quantities are installed quantities, excluding spares. Prices, stock and shipping are not included.

Profile	Installed switches N	Pair extensions	Base M2x6 screws	Printed enclosure pieces
Modular 4	4	0	4	3
Modular 6	6	1	6	5
Modular 8	8	2	8	7
Modular 10	10	3	10	9
Main 10, fixed split chassis	10	Not applicable	8	4

Use **N** below for the chosen installed switch count. **AR** means as required. Optional feature quantities apply only when that feature is fitted. Parts inside boards are not itemized. The Nano mounts directly on integral posts in the electronics base. Electrical reference designators match the Nano wiring diagrams and KiCad schematic.

1. Required control electronics

ID / references	Fitted qty	Part and specification
E01 / U1	1 piece	**Arduino Nano ESP32 ABX00083**. ESP32-S3, 16 MB flash; 43.18 x 17.78 mm PCB; without headers.
E02 / U2	1 piece	**Waveshare 24382**. 1.69inch LCD Module; ST7789V2; 240 x 280 native / 280 x 240 landscape; 31.5 x 39 mm PCB.
E03 / SW11	1 piece	**Rotary encoder with push switch**. Passive quadrature, 20 detents/revolution; normally-open click; 15 x 15 mm body.
E04 / SW1-SW10	N piece	**Momentary footswitch**. Normally open, non-latching; 12 mm mounting thread; 30 x 13 x 15 mm body; shaft offset 9 mm from short end.
E05 / R7	1 piece	**10 kOhm ladder pull-up**. 1% metal-film, through-hole; 0.25 W procurement choice.
E06 / R8	1 piece	**20 kOhm idle resistor**. 1% metal-film, through-hole; 0.25 W procurement choice.
E07 / R9-R18	N piece	**1 kOhm ladder sections**. 1% metal-film, through-hole; 0.25 W procurement choice.
E08 / C5	1 piece	**100 nF ladder filter capacitor**. Ceramic, through-hole; >=10 V rating (50 V is suitable).
E16 / R50	1 piece	**USB VBUS sense upper resistor**. 100 kOhm, 1%, 0.25 W.
E17 / R51	1 piece	**USB VBUS sense lower resistor**. 150 kOhm, 1%, 0.25 W.

E16/E17 form the required USB VBUS divider at A4/GPIO11. All discrete resistors are 1% metal-film; 0.25 W through-hole is a practical procurement choice. Higher precision is optional. C5 100 nF belongs to the switch ladder; C6 100 nF bypasses the 74HCT125 supply. Four 1 kOhm sections are in the base; each two-switch extension adds two. No resistor terminates the final tap.

2. Feature-dependent electronics

Two passive pedal inputs

ID / references	Fitted qty	Part and specification
E09 / J2	1 piece	**Amphenol ACJS-MHDC**. One dual 6.35 mm switched TRS socket housing; fitted horizontally in chassis.
E10 / R21, R31	2 piece	**10 kOhm pedal bias resistors**. 1% metal-film, through-hole; 0.25 W.
E11 / R20, R30	2 piece	**5.1 kOhm pedal reference resistors**. 1% metal-film, through-hole; 0.25 W.

Fit J2 once for one or both pedal inputs. Each populated input adds one 10 kOhm bias resistor and one 5.1 kOhm reference resistor. The table fits both inputs. Across the complete two-input build there are **three 10 kOhm resistors total**, including R7 on page 1. The external pedals are passive; no active interface or extra ADC board is fitted.

Optional 9 V power input

ID / references	Fitted qty	Part and specification
E12 / J1	1 piece	**Barrel power socket**. 5.5 x 2.1 mm; external thread diameter 8 mm, length 8 mm; 4 mm flange/body + 6 mm terminals = 18 mm total.
E13 / U3	1 piece	**Pololu S13V10F5 #4083**. 5 V, nominal 1 A step-up/step-down regulator; about 8.9 x 12.1 mm.
E14 / C1	1 piece	**47 uF input capacitor**. Polarized electrolytic; 35 V minimum; compact radial package.

J1, U3 and C1 are omitted as a group in a USB-only electrical build. The existing chassis still contains their openings and mount features. Pololu 4083 is a **buck-boost** module, not just a step-down regulator; Kamami's 1178565 listing refers to this part. Nano VIN takes the positive 9 V sleeve directly; Pololu VOUT supplies the LED rail only. The converter 1 A rating is subject to thermal conditions. Do not feed its 5 V to Nano VIN or USB VBUS. [S3]

Optional LED feedback

ID / references	Fitted qty	Part and specification
E15 / LED1-LED10	N piece	**Pololu 2535**. 5 mm diffused through-hole RGB pixel with integrated WS2811; 5 V; four leads DIN/5V/GND/DOUT.

Pololu 2535 is sold as a **10-pack**: buy one pack for any single 4-10 switch build and fit N LEDs. The chain uses LEDs connected to +5 V, GND and data wiring. C1 is the converter input capacitor. Direct 3.3 V data into 5 V LEDs requires verification with the actual batch and harness. [S4]

Buffered TRS MIDI OUT

ID / references	Fitted qty	Part and specification
E18 / U4	1 piece	**74HCT125 quad buffer**. 5 V TTL-compatible quad buffer; DIP-14 or SOIC-14; gates 1 and 2 used with inputs 2/5 joined and /OE 1/4 grounded; unused inputs 9/12 and /OE 10/13 to GND.
E19 / D1, D2	2 piece	**1N5819 Schottky diodes**. 40 V, 1 A Schottky rectifier, DO-41; source-isolation diode OR.
E20 / R52	1 piece	**10 kOhm UART idle pull-up**. 1% metal-film, through-hole; 0.25 W. Terminals 3V3 to MIDI_OUT_TX.
E21 / R53, R55	2 piece	**220 Ohm MIDI current-loop resistor**. 1% metal-film, through-hole; 0.25 W. Terminals MIDI_5V to J3_RING / J4_RING.
E22 / R54, R56	2 piece	**220 Ohm MIDI current-loop resistor**. 1% metal-film, through-hole; 0.25 W. Terminals TX_GATE1 to J3_TIP / TX_GATE2 to J4_TIP.
E23 / C6	1 piece	**100 nF buffer bypass capacitor**. 100 nF X7R ceramic, >=10 V, through-hole or 1206. Terminals MIDI_5V to GND.
E24 / J3, J4	2 piece	**3.5 mm TRS MIDI OUT sockets**. 3.5 mm stereo (TRS) panel-mount socket; wired MIDI TRS Type A.

U4 gates 1 and 2 share one UART input and each drive one 3.5 mm TRS Type A current loop: tip = DIN pin 5, ring = DIN pin 4, sleeve = DIN pin 2 shield to GND. UART1 TX is routed to D0/GPIO44; D1/GPIO43 stays unconnected to avoid boot-ROM output. D1/D2 diode-OR USB VBUS and converter +5V to `MIDI_5V`; this supply isolation needs electrical bench verification.

3. Mounting hardware

ID / references	Fitted qty	Part and specification
H01	See p.1	**Base screws**. M2 x 6 mm countersunk.
H02	4 piece	**Display screws**. M2 x 5 mm countersunk.
H03	4 piece	**Nano PCB screws**. M1.6 x 5 mm pan head.
H04	N set	**Footswitch mounting hardware**. Matching nuts and washer set supplied for each selected footswitch.
H05	1 piece	**Encoder knob**. 15 mm diameter x 5 mm height; bore matched to selected shaft.
H06	1 set	**Encoder panel hardware**. Matching retaining nut and washer.
H07	1 set	**TRS panel hardware**. Two nickel nuts and two black fiber washers for ACJS-MHDC.
H08	1 piece	**DC socket retaining nut**. Matching 8 mm thread; 2 mm thick.

Avoid double counting included hardware. H04 is one complete footswitch hardware set per switch; the reference drawing shows two nuts and one washer. H06 is the encoder's nut/washer set. H07 is J2's two nickel nuts and two fiber washers, shown in the attached manufacturer drawing. H08 is J1's 2 mm retaining nut. Check the seller's package contents before ordering any of these separately.

H01 quantity: main enclosure = eight. Modular = four on the electronics base plus two per added pair. The cap has an integral floor and no base screws. Use countersunk heads for H01 and H02.

Display fit: four M2x5 screws pass through the printed hollow mounts into the factory blind studs. Nominal engagement is 1.39 mm; confirm head seating and usable stud depth on the delivered display. Do not substitute the base's M2x6 screws for the display screws.

Nano mounting: four H03 M1.6x5 pan-head screws attach the board component-side up directly to four integral base posts. Posts have 6 mm shafts, 8 mm root envelopes, 45-degree root transitions and 4.3 mm-deep pilots. The USB-side pair has rear flats for pocket clearance; chamfered channels under the opposite pair preserve the vent. Check screw fit and USB cable seating before tightening.

Control fit remains to confirm: no exact footswitch or encoder manufacturer SKU is selected. Match the specified switch body and shaft offset, actuator height, encoder bushing length through the 3 mm roof, shaft form and knob bore. The encoder's 6 mm shaft and 8 mm body height are CAD assumptions. These checks determine the final orderable parts, not a change to the enclosure geometry.

4. Printed enclosure and assembly materials

Choose **one enclosure family**, not both. Main 10 is 350 mm wide; modular widths are 164.85 / 229.85 / 294.85 / 359.85 mm for 4 / 6 / 8 / 10 switches. The ten-switch modular enclosure is wider than the fixed main enclosure.

ID / print item	Mod 4	Mod 6	Mod 8	Mod 10	Main 10
P01 Electronics upper shell	1	1	1	1	1
P02 Electronics base	1	1	1	1	1
P03 Main outer upper shell	0	0	0	0	1
P04 Main outer base	0	0	0	0	1
P05 Repeatable pair upper	0	0	1	2	0
P06 Cap-adjacent pair upper	0	1	1	1	0
P07 Universal pair base	0	1	2	3	0
P08 Integral-floor end cap	1	1	1	1	0

Preferred Left print selection

Main uses one each of **shell-left, shell-right, base-left, base-right** in design/fusion/mirrored/main/. Modular uses the electronics shell/base and one **right-cap**; every extension uses the universal **pair-base**. Upper-shell labeling depends on position:

Modular total	Repeatable upper(s)	Cap-adjacent upper
4	None	None
6	None	pair-shell-at-right-cap-C
8	pair-shell-C	pair-shell-at-right-cap-D
10	pair-shell-C, pair-shell-D	pair-shell-at-right-cap-E

Use the matching files in design/fusion/mirrored/modular/. Right builds use design/fusion/print/ and design/fusion/modular/, a left end cap and their matching labeled uppers. The print guides are authoritative for filenames and orientations. Print only designated shell/base/cap solids, not modeled electronics or reference hardware.

Stock and consumables

ID / references	Fitted qty	Part and specification
M01	AR mass	**PETG filament**. For the selected supplied enclosure print files.
M02	AR set	**PETG-compatible adhesive**. For upper joint chain and removable-base joint chain.
M03	4 piece	**Non-slip feet / clearance pads**. Maintain >=3 mm open space beneath floor vents.

Current supplied profiles use Prusa MK3S+, PETG, a 0.6 mm nozzle and 0.3 mm layers. A 1 kg filament spool is a stock allowance, not an estimate of consumed mass. Slice the actual selected parts with supports, brim and fit coupons. Mounts, converter spring clips, positioning ribs, support pins and glue lands are integral printed features, not separate purchased parts.

5. Wiring and external accessories

Harness materials

ID / references	Fitted qty	Part and specification
W01	1 piece	**LCD cable**. GH1.25, 8-pin mating cable; verify orientation and signal labels.
W02	AR length	**Signal hookup wire**. Stranded, insulated 26-28 AWG; assorted purpose colors.
W03	AR length	**Supply/return wire**. Stranded, insulated 22-24 AWG.
W04	1 set	**Carrier perfboard and U4 socket**. 80.5 x 20 x 1.5 mm perfboard, 6 x 28 holes at 2.5 mm pitch, four 3 mm mounting holes on 76 x 16 mm, four 2 x 3.5 mm pads at each end, plus one DIP-14 socket; carries the U1-side passives, ladder top segment and TRS MIDI OUT stage.
W05	AR set	**Insulation and strain relief**. Heat-shrink, small ties/anchors and insulating sleeves.

Plan about **5 m total of assorted signal wire** and **2 m total supply/return wire** for stock. These are procurement allowances, not measured cut lengths. Keep purpose colors consistent, insulate solder joints, and leave service slack between upper ports and base electronics. Connector families for optional detachable section harnesses and a carrier-board outline are not finalized. Direct solder wiring is the documented electrical baseline; do not assume a custom PCB exists.

Equipment outside the enclosure

ID / references	Fitted qty	Part and specification
A01	1 piece	**USB-C data cable**. Suitable USB host/source end.
A02	1 piece	**USB 5 V source**. Capacity for measured controller and any USB-powered LED load.
A03	1 piece	**9 V DC adapter / pedalboard output**. Center-negative 5.5 x 2.1 mm; >=1 A procurement allowance.
A04	2 piece	**Passive continuous pedal**. Variable resistor or potentiometer, up to 10 kOhm.
A05	1 piece	**BLE MIDI DIN dongle**. Central-role support and third-party peripheral compatibility.

A04 allows up to two passive pedals; zero or one is also valid. A05 is needed for a direct HX Stomp DIN connection, not for connection to an iPad BLE MIDI application. The dongle must be able to act as a central for a third-party peripheral; interoperability has not been bench qualified. HX Stomp, iPad, printer, tools and their own power supplies are not included in the controller BOM.

USB and 9 V sources are alternative power options. Choose each supply for the complete measured load. Ten Pololu 2535 LEDs draw about 0.5 A at full white before the controller/display load. A higher-current input adapter does not increase the converter's 5 V output/thermal rating. Confirm how the LED 5 V branch is powered in a USB-only build; do not join independent positive supply outputs. [S3, S4]

6. Procurement notes and traceability

Before placing the order

Item	Required check
Arduino Nano ESP32	ABX00083 without headers; direct wiring, components up on integral base posts. Calibrate analog inputs after assembly.
Waveshare 24382	Non-touch 1.69inch LCD Module; verify rear M2 studs and GH1.25 8-pin cable. Do not substitute a similar touch display.
Footswitch / encoder	Exact supplier SKUs remain open; match the dimensional and contact requirements on pages 1 and 3.
J1 power socket	Match 8 mm thread, 8 mm threaded length, 18 mm total envelope and 2 mm nut; flange and nut outside diameters still require fit confirmation.
MCU screws	Four M1.6x5 pan head; verify printed pilot engagement.
Bundled hardware	Check cable, header, nuts and washers already supplied with the boards and controls. Do not buy duplicate sets.

Quantity and source checks

The maximum electrical BOM/netlist contains **57 component instances**: three purchased boards, two socket assemblies, one encoder, ten footswitches, twenty-three resistors, three capacitors, two Schottky diodes, one logic buffer, two TRS MIDI sockets and ten LEDs. This includes the two required USB VBUS sense resistors R50/R51, both pedals, 9 V power and optional LEDs. The similarly sized CSV line count is coincidental: CSV rows also include grouped hardware, printed parts and external accessories.

For all optional circuits fitted, electrical instance totals are **39 / 45 / 51 / 57** at 4 / 6 / 8 / 10 switches. Omitting LEDs subtracts N; omitting 9 V input subtracts three (J1/U3/C1). Omitting the complete pedal interface subtracts five (J2 and four resistors). Printed/hardware quantities are separate. The BOM generator checks all 57 reference designators against both electrical exports. It verifies eight base and four display screws against the CAD body inventory, and four M1.6x5 PCB screws against the direct Nano mount specification.

References

[S1] Arduino Nano ESP32 documentation: board identity and manufacturer dimensions. **[S2]** Waveshare 24382 documentation: module, connector and mechanical specification.

[S3] Pololu S13V10F5 #4083: regulator, current/thermal limits and input transient capacitor guidance. **[S4]** Pololu 2535: 10-pack, supply and LED package.

Local basis: design/electrical/bom.csv and netlist.json; design/fusion/mirrored/layout.json; enclosure assembly guides; docs/hardware.md and power.md; and design/fusion/datasheets/double-trs-ACJS-MHDC.pdf. The manufacturer's TRS drawing shows the included nuts/washers and optional PCB support hole.

Prepared 30 October 2026 for SIGMIDI v0.1. The Nano firmware, harness and integral components-up mounting are specified here. Electronics bases carry four reinforced Nano mounting posts. Values and CAD fit describe the current prototype; purchased-part fit, direct LED drive, power/thermal performance and physical assembly still require bench checks. Edit docs/bom-items.json and rerun scripts/build_bom.py to refresh the PDF, Markdown and shopping CSV.